MA2104(3)

Test 2

Tuesday, 20/10/2026

Time: 40 minutes Total marks: 15

1. Given that

$$\coth z = 1 + 2i$$

show that

(a) (3 points) $e^{2z} = 1 - i$

(b) (3 points) $z = \frac{1}{4}\ln(2) + i\left(-\frac{\pi}{8} + k\pi\right), \quad k \in \mathbb{Z}$

2. (4 points) Evaluate

$$\int_{\gamma(0;2)} f(z)\,\mathrm{d}z, \quad \text{where} \quad f(z) = \overline{z}.$$

3. (5 points) Prove that $u(x, y) = e^{-2y}\left[(x^2 - y^2)\cos(2x) - 2xy\sin(2x)\right]$ is harmonic.

[6 + 4 + 5 = 15 marks]